Conditional Lift Descent for Finite Observer-Relative Obstruction Systems

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> **Non-claims:** This paper does not prove universal structural growth, universal lift-completeness of EdgeSplit/CycleHub, observer-independent descent, Gödel incompleteness, physical autopoiesis, or semantic truth from obstruction mass. Positive Φ licenses a lift only after internal repair exhaustion and verified post-lift descent inside a declared finite candidate class.

Abstract

We define an observer-relative lift constructor for finite obstruction systems with restricted internal repair. An operational state consists of a finite carrier P, vertex data $s \in C^0(P)$, an observed edge field $\omega \in C^1(P)$, a declared observer functional $J_{\mathcal{O}}$, and an admissible repair class $\mathrm{Adm}(P)$. The residual obstruction is not the unrestricted coboundary $d_0 s$; it is the post-repair residual

$$h = \omega - d_0 f^*, \qquad f^* \in \arg\min_{f \in \mathrm{Adm}(P)} J_{\mathcal{O}}(\omega - d_0 f).$$

Thus positive obstruction requires either a non-exact observed field, a restricted repair class, or both. A structural lift is a finite local extension of the carrier along $\mathrm{supp}(h)$, followed by a post-lift optimization of the extended section and observed field. We define admissible lift candidates, a minimal generator class generated by EdgeSplit and CycleHub operations, and a cost-penalized lift selection rule.

The main result is conditional: under a fixed observer, exhausted internal repair, and a verified lift-complete candidate class containing a Φ-reducing extension, the selected lift strictly decreases the declared observer-relative obstruction. If no such candidate is verified, the correct terminal state is refusal, budget unknown, or lift-class incompleteness rather than structural expansion. The construction is therefore a finite operational discipline for licensed lift, not a universal growth theorem.

> **v013 Response to Red Team (Round 10) — Summary of Changes**

Goal / Attack	Status	Response
R8.1: Soften Gödel wording	**Fixed**	§18.6 toned down "exact/direct mapping" to "operationally analogous / modeled here as" to stay within mathematical discipline.
R8.2: Simplify Definition 18.1	**Fixed**	§18.2 simplified to a direct property of strict descent bounds.
R8.3: Weaken "Impossibility"	**Fixed**	§18.5 corrected "mathematical impossibility" to strict contradiction of the lift-completeness assumption.
R8.4: Expand Code Correspondence	**Fixed**	§15.2 expanded to cover exhaustion condition and explicit candidate lift generation routines.

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16. Structural Foundations (v004, revised in v005–v006)

Sections 16.1, 16.3–16.8 are unchanged from v005. Only §16.2 was revised in v006.

16.2 Representation Conjecture (v015)

Conjecture 16.1 (Representation Conjecture). Let S be a finite-state system equipped with edge-locality, an observable Φ_S, and a bounded convergence law halting with internal operators exhausted. Then S admits a representation as a lift-capable system (P, s, ω, H, C) **up to Φ-observational equivalence**, subject explicitly to the Disjoint Observer Consistency Theorem rendering mapping choices to **Conditional Minimality**.

This conjecture is not used in the proof of Theorem 18.1.

17. Conditional Categorical Lift Principle (v015)

Unchanged from v006.

17.5 Observer-Relative Metric Structure

All quantities involving obstruction Φ depend explicitly on a chosen observer model $\mathcal{O} = (\mathcal{A}, \mathcal{M}, \mathcal{J})$. The specific ℓ^1 metric structure derived throughout this paper arises canonically only for replica-extensive, decomposable observers acting on structurally independent edge stalks. Other observers (e.g., max-bottleneck, correlated ℓ^2) induce alternative obstruction geometries with completely different optimal configurations. All subsequent iterative dynamics and metric descent bounds are fundamentally defined relative to this initial formal operational choice.

18. The Structural Lift Constructor Λ_h (v015)

The goal is not "growth happens somehow." The goal is:

$$\Phi > 0 \wedge \text{no admissible descent in } P \quad \Longrightarrow \quad \text{construct } \Lambda_h \text{ and extend } P \text{ to } P'$$

Within the stated observer and repair boundaries, this shifts structural augmentation from an arbitrary choice to a conditional construction driven by the irreducible inconsistency h.

18.1 Residual Inconsistency and Exhaustion

An operational state is (P, s, ω), where $s \in C^0(P)$ records the current vertex-level section data and $\omega \in C^1(P)$ is the observed edge field supplied by the observer. No exactness axiom is imposed; in general $\omega \notin \mathrm{im}(d_0)$ and need not equal $d_0 s$.

Let $\mathrm{Adm}(P) \subseteq C^0(P)$ be the admissible internal repair class. The best possible internal repair is:

$$f^* \in \arg \min_{f \in \mathrm{Adm}(P)} J_{\mathcal{O}}(\omega - d_0 f)$$

and the **residual inconsistency** is:

$$h = \omega - d_0 f^*.$$

Thus:

$$\Phi_{\mathcal{O}}(P, \omega; \mathrm{Adm}) = J_{\mathcal{O}}(h).$$

The minimum is attained for finite P in the standard unrestricted finite-dimensional subspace case, or whenever the admissible repair image is proximinal for the chosen norm (compact admissible sets are sufficient). Otherwise the definition must be written with inf and approximate residuals. If $\mathrm{Adm}(P) = C^0(P)$

and $\omega \in \mathrm{im}(d_0)$, then $\Phi_{\mathcal{O}} = 0$. Hence positive obstruction requires either a genuinely non-exact observed field, a restricted repair class, or both.

In this paper, we restrict exclusively to the replica-extensive observer class mapping disjoint measurement bounds, hence solving under the declared scalar graph specialization:

$$\Phi = \|h\|_1$$

18.2 Support of the irreducible obstruction (Definition 18.3)

Let $h \in C^1(P)$ be the irreducible residual. Define its support by:

$$\mathrm{supp}(h) := \{e \in E(P) : h(e) \neq 0\}$$

Define the vertex support:

$$V(h) := \{v \in V(P) : v \text{ is incident to some } e \in \mathrm{supp}(h)\}$$

This is the minimal region where growth is allowed to occur.

18.3 Admissible lift (Definition 18.4)

An admissible lift over P relative to h is a finite extension $\Lambda = (V_\Lambda, E_\Lambda)$ such that the extended carrier $P' := P \cup \Lambda$ satisfies:

1. **Locality**: $V(\Lambda) \subseteq N_r(V(h))$ for some fixed radius r.
2. **Nontriviality**: $|V(P')| > |V(P)| \quad \text{or} \quad |E(P')| > |E(P)|$.
3. **Compatibility**: The extension preserves the admissible carrier class: poset/graph validity, incidence consistency, and whatever local orientation rules P already obeys.
4. **Rank increase along obstruction support**: If d_0' is the coboundary operator on P', then

   $$\dim \mathrm{im}(d_0'|_{V(h)\cup V_\Lambda}) > \dim \mathrm{im}(d_0|_{V(h)})$$

   Rank increase is only an admissibility filter. It is not a descent certificate. A lift passing this rank test must still pass the post-lift optimization check $\Phi(P \cup \Lambda) < \Phi(P)$.

This makes "growth" precise: it is not arbitrary mutation, but local enlargement of representational degrees of freedom along the support of h.

18.4 Post-lift section update (Definition 18.6)

For each admissible lift Λ, define the post-lift pair by:

$$(s^\sharp_\Lambda, \omega^\sharp_\Lambda) \in \arg \min_{(\hat{s},\hat{\omega})\in \mathrm{Ext}(P,s,\omega,\Lambda)} J_{\mathcal{O}}(\hat{\omega} - d'_0\hat{s})$$

where $\mathrm{Ext}(P, s, \omega, \Lambda)$ means admissible post-lift extensions that preserve (s, ω) outside a bounded neighborhood of $V(\Lambda)$. This keeps locality while allowing necessary local relaxation near the lift region.

This means growth is constructed in two stages: 1. choose structure Λ, 2. solve the best local section extension on that new structure.

18.5 Lift operator (Definition 18.5)

Given exhausted state (P, s, ω) with residual $h \neq 0$, define the lift operator:

$$\mathcal{L}_h : (P, s, \omega) \mapsto (P', s', \omega')$$

by the rule $P' = P \cup \Lambda^*_h$, where:

$$\Lambda^*_h = \arg \min_{\Lambda\in\mathcal{L}(P,h)} \Big(\Phi_{\mathcal{O}}(P \cup \Lambda, \omega^\sharp_\Lambda; \mathrm{Adm}') + \lambda|\Lambda|\Big)$$

Here: * $\mathcal{L}(P, h)$ is the class of admissible lifts over P relative to h, * $|\Lambda|$ is the structural cost: $|\Lambda| := \alpha|V_\Lambda| + \beta|E_\Lambda|$, * $(s^\sharp_\Lambda, \omega^\sharp_\Lambda)$ is the optimal post-lift representative on $P \cup \Lambda$, * Adm' is the post-lift admissible repair class.

18.6 Minimal Lift Class $\mathcal{L}_{\min}$

We restrict $\mathcal{L}(P, h)$ to a proposed minimal lift basis: only the lift primitives currently claimed to produce strict Φ-descent when internal repair is exhausted.

Definition (Minimal Lift Class) $\mathcal{L}_{\min}(P, h)$ is the finite-composition closure of exactly two primitive generators with distinct roles:

1. **CycleHub** *(candidate resolver)*: For a minimal cycle $C = (v_1, \ldots, v_k, v_1) \subseteq \mathrm{supp}(h)$, introduce one new hub vertex c_C joined to the cycle. This is a candidate resolver only after the lifted incidence/repair class is specified; adding ordinary graph edges alone does not automatically kill a quotient obstruction class.
2. **EdgeSplit** *(refinement)*: For an obstructed edge $e = (u, v)$, introduce one new latent vertex m_e and refine by $(u, v) \rightsquigarrow (u, m_e), (m_e, v)$. This preserves $\beta_1 = |E| - |V| + 1$ for connected graphs and therefore cannot be assumed to resolve a purely cycle-supported quotient obstruction by homology alone.

(Note: EdgeSplit preserves cycle rank because $|E|+1-(|V|+1)+1 = |E|-|V|+1 = \beta_1$*. It may change support, cost, or admissible repairs, but strict* Φ*-descent*

still has to be verified by the post-lift optimization problem. CycleHub is not automatically resolving without the corresponding higher-cell or incidence data.)

Thus:

$$\mathcal{L}_{\min} = \langle \text{EdgeSplit}, \text{CycleHub} \rangle_{\text{finite composition}}$$

18.7 Minimal Lift Admissibility and Completeness

A candidate $\Lambda \in \mathcal{L}_{\min}(P, h)$ must pass a strict computational rank-boost test locally. If $\operatorname{rank}\big(\operatorname{im}(d_0')|_{V(h)\cup V_\Lambda}\big) \not> \operatorname{rank}\big(\operatorname{im}(d_0)|_{V(h)}\big)$, it is not admissible under this search rule.

Definition 18.7 (Minimal Lift-Completeness) $\mathcal{L}_{\min}$ is conjecturally lift-complete for the finite edge-local additive class: every exhausted residual h admitting a local finite Φ-reducing extension admits one explicitly generated by finite composition of EdgeSplit and CycleHub lifts.

Minimal Search Rule: 1. **Tier 1:** All single `CycleHub` lifts on minimal cycles in $\operatorname{supp}(h)$ 2. **Tier 2:** `CycleHub` + `EdgeSplit` compositions 3. **Stop** at the first tier yielding any Φ-reducing candidate; apply $\arg\min$ within that tier only.

Minimal Cost Objective:

$$\Lambda_h^* = \arg\min_{\Lambda \in \mathcal{L}_{\min}(P,h)} \Big(\Phi_{\mathcal{O}}(P \cup \Lambda, \omega_\Lambda^\sharp; \text{Adm}') + \lambda|\Lambda| \Big) \quad \text{where } |\Lambda| = |V_\Lambda| + |E_\Lambda|$$

18.8 Theorem 18.1 — Conditional Lift Descent Under Verified Lift-Completeness

Assumptions: 1. A fixed observer $\mathcal{O}$ selecting the ℓ^1 metric (RE case). 2. A finite state (P, s, ω) with non-zero residual $h = \omega - d_0 f^*$ evaluating $\Phi_{\mathcal{O}} = \|h\|_1 > 0$. 3. Restricted to an admissible internal update class strictly exhausted inside P. 4. A verified lift-complete extension class $\mathcal{L}(P, h)$.

The lift-completeness assumption is substantive. It is not proved in this paper. The theorem is therefore a conditional descent theorem: once a finite candidate class has been verified to contain a Φ-reducing lift for the exhausted residual h, the selected lift decreases the declared observer functional.

Conclusion: The explicitly generated lift operator $\mathcal{L}_h(P, s, \omega) = (P', s', \omega')$ guarantees:

$$\Phi_{\mathcal{O}}(P', s', \omega') < \Phi_{\mathcal{O}}(P, s, \omega).$$

Crucial Boundary Constraint: This geometric descent guarantee is generally **not invariant** under a change of observer. Descent flows bounded favorably in ℓ^1 topological tracking will routinely expand and corrupt concurrent metrics bounded entirely under alternative ℓ^∞ or ℓ^2 norms on the equivalent configuration.

Interpretation: Under a fixed observer, admissible update class, and verified lift-complete extension class, failure of internal descent justifies formal geometric extension.

18.9 The Explicit Operational Law

The operational law is conditional:

$$h \neq 0 \wedge \neg\exists u \in \mathrm{Adm}(P) : \Phi(u(P,s,\omega)) < \Phi(P,s,\omega) \wedge \exists \Lambda \in \mathcal{L}(P,h) : \Phi(P \cup \Lambda, \omega^\sharp_\Lambda; \mathrm{Adm}') < \Phi(P,\omega;\mathrm{Adm})$$

$$\Longrightarrow (P,s,\omega)\mathcal{L}_h(P',s',\omega')$$

with $P' = P \cup \Lambda^*_h$, $|P'| > |P|$, and strict observer-relative descent.

If no such Λ is found within the declared search class, the correct terminal status is not lift, but one of:

$$\mathsf{NoLiftFound}, \quad \mathsf{BudgetUnknown}, \quad \mathsf{LiftClassIncomplete}, \quad \mathsf{Refused}.$$

18.10 Structural Analogy: Exhaustion and Extension

This algorithm design logic functions as a strict computational analogy bridging logical exhaustion inside bounded configurations:

- **Internal exhaustion:** The declared repair class fails to reduce the residual h inside the current carrier P.
- **Structural extension:** A verified lift enlarges the carrier and repair image so that the post-lift optimization problem admits strict observer-relative descent.

This is an operational analogy to formal extension after exhaustion. No claim about Gödel incompleteness, undecidability, or formal arithmetic is proved here.

15.2 Code Correspondence Table (v012 Context)

This is the minimal OS-aligned structure hiding inside the math operator $\mathcal{L}_h$.

```
enum LiftPrimitive {
    EdgeSplit { edge: EdgeId },
    CycleHub { cycle: Vec<VertexId> },
}

trait LiftGenerator {
    fn generate_minimal_lifts(&self, p: &Carrier, h: &Residual) -> Vec<LiftPrimitive>;
}
```

Paper 005 Entity	Code Entity	Lines	Notes
$h = \omega - d_0 f^*$	`compute_phi_total`	—	Total phi calculates exactly $\|h\|_1$ after dual internal loops
Exhaustion Condition	`run_internal_repairs` bounds		Tracks stagnant step/delta changes indicating f^* failure mode
$\Lambda \in \mathcal{L}(P, h)$	`LiftCandidate`	—	Emits specific local graph expansions bridging the support boundary
$\mathcal{L}_h$ Application	`apply_structural_lift`	947-952	Injects latent variable exactly corresponding to selected Λ_h^*
Cost Parameter Rollback	`revert_lift` procedure	1080-1083	Reverting lift explicitly safeguards the $\lambda\|\Lambda\|$ cost threshold constraint

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Changelog

v016 (May 2026)

- Cross-referenced Paper 000 v004 to explicitly bind the structural lift constructor to the general theory's contextual congruence and observer passivity theorems.

v015 (April 2026)

- Corrected the exactness collapse: raw obstruction is an observed edge field $\omega \in C^1(P)$, not the unrestricted coboundary $d_0 s$.
- Made admissible repair classes explicit and scoped minimizer existence to finite closed repair classes.
- Weakened lift claims: CycleHub and EdgeSplit are candidate generators whose descent effect must be verified by the post-lift optimization problem.

v014 (April 2026)

- Red Team synthesis: Explicitly anchored representation boundaries into the **Conditional Minimality** framework defined by the Disjoint Observer Consistency Theorem.

v013 (April 2026)

- Red Team Round 10: Compressed $\mathcal{L}(P, h)$ to Minimal Lift Class $\mathcal{L}_{\min}$.
  - Established two primitive basis generators: `EdgeSplit` and `CycleHub`.
  - Defined strict structural admissibility rule relying strictly on local $\mathrm{im}(d_0)$ rank inflation.
  - Eliminated arbitrary topological structural mutations in favor of minimal computationally bounded searches.

v012 (April 2026)

- Red Team Round 9 final constructive bounds:
  - Defined exact admissible lift $\mathcal{L}(P, h)$, lift operator $\mathcal{L}_h$, and post-lift section update.
  - Formally derived structural lift definitions (Types I, II, III minimal topologies).
  - Mapped operator strictly to explicit functional Rust traits.

v011 (April 2026)

- Red Team Round 8 final editorial and mathematical tidying:
  - Definition 18.1 (Lift-Completeness) significantly simplified to plain strict-descent boundaries.
  - Softened remaining Gödel analogies avoiding over-reaching rigorous proof bounds.
  - Code equivalence table fully expanded to capture procedural generator bounds.

v010 (April 2026)

- Red Team Round 7 finalized logic bounds:
  - Universality deliberately constrained to strictly finite, edge-local additive systems operating under lift-complete bounds.

(For legacy details spanning v001-v009, reference canon `.archives`)